

Profile Card UI

Assignment 3 (Brief) — Profile Card with Column, Row, Container, CircleAvatar & Custom Theme Colors

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Environment: Flutter 3.x / Dart 3.x

Code: https://github.com/SoumitraDeshpande11/assignments/tree/assignments_4

1. Objective

Build a Flutter profile card screen using the core layout widgets — Column, Row, Container, CircleAvatar, Text, and Icon — styled with custom theme colors.

2. Requirements Checklist

Requirement	Where it lives
Column	Main card layout — avatar, name, stats, contacts stacked vertically
Row	Location line, the three stat boxes, contact rows, and the action buttons
Container	The card itself, each stat box, and the icon chips — all with custom padding, color, and rounded corners
CircleAvatar	Profile picture placeholder with initials "SD"
Text	Name, role, stats, and contact details in varying sizes/weights
Icon	Location, email, phone, website, and button icons
Custom theme colors	kPrimary, kAccent, kSurface constants feeding ColorScheme.fromSeed and the card styling

3. How the Layout Works

The screen is a single centered card built from nested layout widgets:

1. A Container with a light surface color and 20px rounded corners forms the card.
2. Inside it, one Column stacks everything top to bottom.
3. Rows handle the horizontal pieces: three equal stat boxes (Expanded inside a Row), contact lines (icon chip + text), and the Follow/Message buttons.

4. Small reusable widgets — `_StatBox` and `_ContactRow` — keep the build method short and consistent.

Custom colors are defined once as constants (`kPrimary` green, `kAccent` light green, `kSurface` off-white) and reused everywhere, so the whole card recolors by changing three lines.

4. Widget Map

MaterialApp (custom green theme)

```
└ Scaffold
  └ AppBar ('Profile')
  └ Center
    └ Container ← the card (rounded, kSurface)
      └ Column
        └ CircleAvatar ← "SD" initials
        └ Text ← name / role / location (Row with Icon)
        └ Row ← _StatBox × 3 (Projects, Followers,
Following)
        └ Divider
        └ _ContactRow ← email / phone / GitHub (Icon in
Container + Text)
        └ Row ← Follow (FilledButton) + Message
(OutlinedButton)
```

5. Source Code

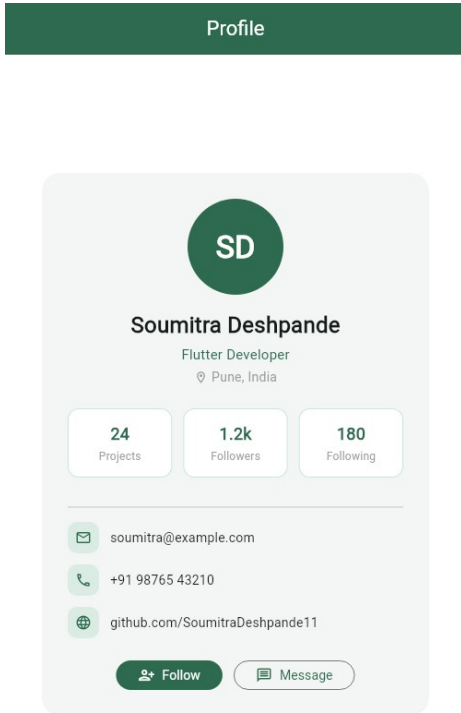
```
lib/main.dart

1 import 'package:flutter/material.dart';
2
3 void main() => runApp(const ProfileApp());
4
5 // Custom theme colors used across the card.
6 const Color kPrimary = Color(0xFF2D6A4F);
7 const Color kAccent = Color(0xFF74C69D);
8 const Color kSurface = Color(0xFFF4F6F5);
9
10 class ProfileApp extends StatelessWidget {
11   const ProfileApp({super.key});
12
13   @override
14   Widget build(BuildContext context) {
15     return MaterialApp(
16       title: 'Profile Card',
17       debugShowCheckedModeBanner: false,
18       theme: ThemeData(
19         useMaterial3: true,
20         scaffoldBackgroundColor: Colors.white,
21         colorScheme: ColorScheme.fromSeed(seedColor: kPrimary),
22       ),
23       home: const ProfileScreen(),
24     );
25   }
26 }
27
28 class ProfileScreen extends StatelessWidget {
29   const ProfileScreen({super.key});
30
31   @override
32   Widget build(BuildContext context) {
33     return Scaffold(
34       appBar: AppBar(
35         title: const Text('Profile'),
36         backgroundColor: kPrimary,
37         foregroundColor: Colors.white,
38       ),
39       body: Center(
40         child: SingleChildScrollView(
41           padding: const EdgeInsets.all(24),
42           child: Container(
43             constraints: const BoxConstraints(maxWidth: 420),
44             decoration: BoxDecoration(
45               color: kSurface,
46               borderRadius: BorderRadius.circular(20),
47             ),
48             padding: const EdgeInsets.all(28),
49             child: Column(
50               mainAxisAlignment: MainAxisAlignment.min,
51               children: [
52                 const CircleAvatar(
53                   radius: 52,
54                   backgroundColor: kPrimary,
55                   child: Text(
56                     'SD',
57                     style: TextStyle(
58                       fontSize: 34,
59                       fontWeight: FontWeight.bold,
```

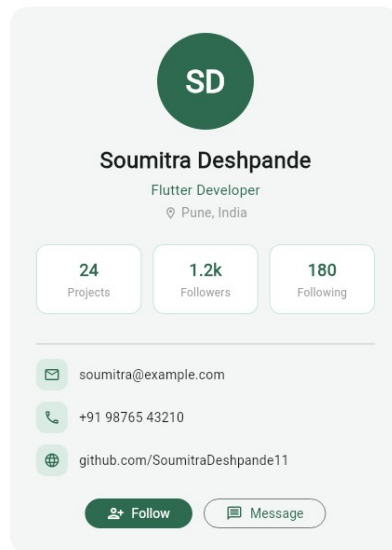
6. App Screenshots

The app was built for the web (`flutter build web`) and captured in Chrome at two window sizes.

Phone width (500px):



Desktop width (1280px):



7. Terminal

Real build-and-run session: `flutter pub get`, `flutter analyze` (no issues), `flutter build web`, then `flutter run -d web-server` to serve the app locally.

```
zsh - assignment_4

$ flutter --version
Flutter 3.47.2 • channel stable • https://github.com/flutter/flutter.git
Framework • revision d3b14c8769 (3 weeks ago) • 2026-08-26 16:07:51 -0700
Engine • hash 1cf1c4773fb941c4c74a7f8bb144a8837596c0f4 (revision a804b26164)
Tools • Dart 3.13.2 • DevTools 2.60.0

$ flutter pub get
Got dependencies!

$ flutter analyze
No issues found! (ran in 3.5s)

$ flutter build web --release
Compiling lib/main.dart for the Web... 25.0s
✓ Built build/web

$ flutter run -d web-server --web-port=8744
Launching lib/main.dart on Web Server in debug mode...
Waiting for connection from debug service on Web Server... 11.7s
lib/main.dart is being served at http://localhost:8744

Flutter run key commands.
r Hot reload.
R Hot restart.
h List all available interactive commands.
d Detach (terminate "flutter run" but leave application running).
c Clear the screen
q Quit (terminate the application on the device).
```

8. Run Instructions

`flutter pub get`

`flutter run` # pick a device, or use `flutter run -d chrome`

9. Possible Extensions

- Load the avatar from a network image with `CircleAvatar(backgroundImage: ...)`.
- Make Follow/Message functional with state (`setState` toggling "Following").
- Animate the card entrance with a fade/slide transition.

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